

MODULE 2

CHAPTER 5 - WONDERS OF SCIENCE = ALBEERT EINSTEIN'S SPEECH AT BERLIN ON 22.08.1930

Dear Present and the Absent,

When you listen to the radio, think how people came into possession of this wonderful tool of communication. The source of all technical achievements is the divine curiosity and the playful instinct of the tinkering and pondering researcher and no less the constructive imagination of the technical inventor. Think of Oersted (Danish physicist who discovered electric current that creates magnetic fields), of Reis (invented electricity propagating through space and transmit voice) who used this effect to generate sound by electromagnetic means, of Bell (Graham Bell, inventor of telephone) who using sensitive contacts with his microphone, first converted sound vibrations into variable electrical currents. Think of Maxwell (who first used his equation to propose that light is an electromagnetic phenomenon) who demonstrated the existence of electric waves by mathematical means and of Hertz (discovered radio waves) who first generated and demonstrated them with the help of sparking. Remember especially Liebens (who made major contributions to electronics and telecommunications like electronic oscillator, television technology, photoelectric effect, synchronized transmitter and receiver etc.) who invented an incomparable sensing organ for electrical vibrations in the electrical valve tube, which also turned out to be an ideally simple instrument for generating electrical oscillations. Remember with gratitude, the army of nameless technicians who simplified the instruments of radio communication and adapted them to mass production in such a way that they became accessible to everyone. And everybody should be ashamed who uses the wonders of science and engineering without thinking and having mentally realized not more of it than a cow realizes of the botany of the plants which it eats with pleasure. Think also about the fact that it is the engineers who make true democracy possible. They facilitate not only the daily work of the people but also make the works of the finest thinkers accessible to the public. The pleasure of these works had recently still been a privilege of the preferred classes. Thus the engineers wake the people from their sleepy bluntness. The radio has to fulfil a special and unique function of international reconciliation. Upto the present day, people got to know each other almost only with the help of the distorting mirror of their own daily press. Radio shows them to each other in their most vivid form and mainly from their amicable side. Thus it will contribute to end the feeling of bilateral strangeness which so easily turns to mistrust and hostility. With this

attitude, look at the results of the creation which this exhibition offers the astonished senses of the visitors.

EXERCISES:

Answer the following questions in two or three sentences:

1. Outline the occasion of Einstein's 1930 Berlin speech?

Ans: Albert Einstein's 1930 Berlin speech took place during the opening of the seventh German radio exhibition, held at the base of the radio tower in Berlin. This event attracted attention as it showcased advancements in radio technology, with Einstein discussing the significance of science and engineering in society.

2. Trace the origin of technical achievements as mentioned in the speech?

Ans: In his 1930 speech in Berlin, Einstein emphasized that the foundation of all technical achievement springs from "divine curiosity" and the inherent drive of researchers to tinker and ponder. He believed that such curiosity, along with the playful exploration of ideas, fuels discoveries and advancements in technology and science. This reflects his view that intellectual curiosity is crucial for innovation and the development of new theories.

3. List the names of the scientists and their contributions towards the invention of radio?

Ans: Oersted, Danish physicist who discovered electric current that creates magnetic fields, Reis who invented electricity propagating through space and transmit voice who further used this effect to generate sound by electromagnetic means, Graham Bell, inventor of telephone) who used sensitive contacts with his microphone, first converted sound vibrations into variable electrical currents, Maxwell who demonstrated the existence of electric waves by mathematical means and of Hertz who discovered radio waves generating and demonstrating them with the help of sparking.

4. Mention how Einstein connects "divine curiosity" of scientists with the practical applications of engineers?

Ans: Einstein connects the "divine curiosity" of scientists with the practical applications of engineers by suggesting that the innate curiosity and playful instinct of researchers drive the pursuit of knowledge, which in turn leads to technological advancements. He believed that this curiosity is the foundation of all technical achievements, highlighting a symbiotic relationship where scientific inquiry inspires engineering solutions that address real-world problems.

5. Comment on the tone and message of the speech?

Ans: Albert Einstein delivered a speech in Berlin in 1930 at a pivotal time in his life and in global history, marked by economic uncertainty and rising political tensions. The tone of this speech, like much of Einstein's public addresses, is reflective, earnest, and hopeful. He focuses on the values of humanity, peace, and intellectual progress, emphasizing the responsibility of individuals to contribute positively to society. The importance of intellectual freedom and creativity, the role of science and technology in improving human life while cautioning against their misuse, a call for peace and understanding, especially significant in the post-World War I climate and during the interwar years had clearly been reflected throughout the speech.

Answer the following questions in a paragraph:

6. Elaborate on Einstein's perspective on the role of curiosity and playful instincts in scientific discovery?

Ans: Einstein viewed curiosity as a fundamental driver of scientific discovery, believing that a deep desire to understand the universe leads to significant breakthroughs. His own relentless passion for knowledge and willingness to challenge conventional wisdom exemplified his belief that playful instincts—like improvisation in thought and exploration—foster creativity in scientific inquiry. He regarded nonconformist thinking and an unconventional approach to learning as essential traits that allow scientists to decipher complex problems and explore new theories, as seen in his work on the unified field theory.

7. Address Einstein's concerns on the potent misuse of scientific knowledge?

Ans: Albert Einstein frequently expressed concerns about the misuse of scientific knowledge, particularly its potential to harm humanity. His worries became especially pronounced after the development and use of nuclear weapons, an invention his theoretical work indirectly influenced. He believed that scientific discoveries, while capable of great progress, could also be exploited for harmful purposes if ethical considerations were not prioritized. His frustration was amplified by the realization that theoretical concepts could outpace societal readiness to handle their implications responsibly. These concerns can be addressed through integrating ethics into scientific education to ensure scientists understand the societal impact of their work, and to inculcate public awareness about the potential risks and benefits of scientific innovations.

8. Sketch Einstein's optimism in perceiving the radio to bridge cultural and geographical divide?

Ans: Einstein viewed the radio as a powerful tool for bridging cultural and geographical divides, believing it had the potential to unite humanity. He saw it as a means to draw the

world closer together, allowing for the sharing of ideas, music, and knowledge across distances. This optimism reflected his belief in the interconnectedness of people and the possibility of fostering understanding and collaboration through technology.

9. Examine the role of radio in international relations?

Ans: Albert Einstein did not specifically address the role of radio in international relations in his works. However, one can infer that radio communication, as a groundbreaking technology, plays a significant role in shaping international relations by facilitating instant communication across borders, influencing diplomacy, and enabling the rapid exchange of information. This can lead to greater global connectivity and understanding, but also to the spread of propaganda and misinformation.

10. Comment on Einstein as an orator?

Ans: Einstein's speeches and lectures often reflected his clear thinking and deep understanding of complex subjects. He had the ability to distill intricate concepts into simpler explanations, particularly for scientific audiences. Einstein was a humble and introspective person, and this extended to his speaking style. He often spoke in a measured, modest tone, focusing more on the content than on engaging theatrics or persuasion. As a German-speaking individual living in various countries, Einstein often gave talks in languages other than his native tongue. This might have posed some challenges in terms of fluency and expressiveness in delivering speeches to non-German audiences. Despite his unassuming delivery, the power of Einstein's speeches often lay in their profound content. His words on peace, science, and humanity carried significant weight because of his intellectual stature and moral authority.

Essay Questions:

11. Examine the collaborative nature of scientific endeavors as envisioned by Einstein?

Ans: Albert Einstein's vision of scientific endeavors reflects a deep appreciation for collaboration and the interconnectedness of ideas within the scientific community. He emphasized that scientific discovery is not a solitary pursuit but rather a collective journey where numerous thinkers contribute to the evolution of knowledge. His interactions with contemporaries, many of whom were unknown in the academic world, underscore the importance he placed on exchanging ideas and fostering discussions.

Einstein believed that engaging with fellow scientists allowed for a more profound exploration of concepts, and he often acknowledged the input and insights of others as critical to his own work. This collaborative dynamic is essential in science, where advancements often rely on building upon previous findings and theories. He exemplified this belief through his own

practices, frequently discussing theories with friends and colleagues, which he described as a necessary aspect of refining his thoughts and hypotheses.

By promoting the idea that scientific progress hinges not only on individual genius but also on the contributions of a broader community, Einstein laid the groundwork for a more inclusive understanding of how knowledge is generated. This view remains foundational in contemporary science, where collaboration across disciplines is vital for addressing complex challenges.

12. Evaluate Einstein's speech as a voice for international cooperation and understanding?

Ans: Albert Einstein's advocacy for international cooperation and understanding is more vividly illustrated in his speech "The Menace of Mass Destruction." than in our speech of pedagogy. In this address, he articulates the profound risks that arise from unchecked technological advancements, particularly in the realm of nuclear weapons. Einstein emphasizes a collective responsibility among nations to foster peace and dialogue rather than engage in conflict. He argues that the threat posed by mass destruction requires a united global response, urging humanity to prioritize diplomacy over militarization.

Einstein's speech can be seen as a clarion call for a new era of international collaboration, where scientific knowledge and ethical considerations guide the decision-making processes of governments. He believes that only through cooperation can nations address existential threats, promoting mutual understanding and respect across borders. By advocating for a collective approach to global challenges, Einstein not only highlights the dangers of isolationism but also offers a constructive vision for a cooperative future built on shared goals and values.

In conclusion, Einstein's speech serves not merely as a warning but as an inspiration for ongoing dialogue and collaboration in the interest of humanity, reflecting his belief in science as a bridge to international understanding.

13. Analyse Einstein's 1930 Berlin speech to examine his views on the ethical responsibilities of scientists and engineers?

Ans: In his 1930 Berlin speech, Einstein emphasized that scientists and engineers have a profound ethical responsibility to consider the implications of their work. He criticized those who exploit the advancements of science and engineering without proper understanding, suggesting that uninformed use diminishes human potential and moral responsibility. Einstein argued that professionals in these fields must engage deeply with

their inventions, recognizing their impact on society and prioritizing ethical considerations over mere technical success.

Einstein highlighted the dual-edged potential of scientific advancements. He acknowledged that science and technology, while having the potential to improve human life, could also lead to destructive consequences if misused. This underscores his belief that scientific knowledge is not inherently moral or immoral; rather, its impact depends on how it is applied. They bear the responsibility of anticipating the consequences of their innovations and ensuring that their work contributes to the betterment of humanity rather than its harm.

Einstein argued that the responsibilities of scientists and engineers extend beyond the act of discovery or invention. He believed they should actively engage in discussions about the societal implications of their work and advocate for its responsible use. This calls for scientists and engineers to act as stewards of their knowledge, participating in policy-making and public education to prevent misuse.

Given the context of the 1930s, Einstein's speech also reflected his concern over the political and social climates of the time, particularly the rise of militarism and nationalism. He foresaw the potential misuse of scientific advancements for warfare, highlighting the ethical dilemmas faced by scientists working under oppressive regimes or in military contexts. Einstein's views underline the need for international cooperation among scientists to address global challenges and prevent the militarization of science.

14. Assess the role of science in shaping human progress and understanding?

Ans: Science has been a pivotal force in shaping human progress and understanding by driving technological advancement, addressing practical issues, and enhancing our collective knowledge. Through scientific inquiry, we have developed tools and technologies that address everyday challenges, improve living standards, and promote efficient resource use. This contributes to practical problem-solving and informed decision-making, which are essential for both individual well-being and societal governance.

Moreover, science extends beyond mere technology; it influences societal constructs by engaging with ethical dilemmas, cultural development, and education. By prompting deeper questions around justice and aesthetics, science enables societies to evolve and adapt in alignment with progressive values, ultimately leading to improved conditions for humanity.

In addition to material progress, science fosters a more profound understanding of our environment, the universe, and ourselves, spurring curiosity and encouraging a quest for knowledge that can bridge gaps in understanding between different cultures and academic

disciplines. As we continue to engage with scientific discovery, we cultivate a foundation for future generations that prioritizes critical thinking and innovation. In summary, science is not just a body of knowledge, but a dynamic catalyst that shapes our world and propels human advancement across various dimensions—technological, ethical, and cultural.